📘 Lecture 28-Switch MAC Table,LAN,VLAN

🧠 1️⃣ MAC Address (Media Access Control Address)

📡 What is a MAC Address?

A **MAC Address** is a **unique physical address** assigned to every network device.

- 📏 **Length:** 48 bits (6 bytes)
- 🧾 Written in **Hexadecimal format**
- 🔥 Permanently stored inside the **Network Interface Card (NIC)**
- Also called **BIA – Burned-In Address**

Example MAC Address:

E8:BA:70:11:28:74

Every device connected to a network has its **own unique MAC address**.

📌 Used for **communication inside a local network (LAN).**

🧩 MAC Address Structure

A MAC address has **two parts**.

Part	Size	Meaning
OUI	First 3 bytes	Identifies manufacturer
Device ID	Last 3 bytes	Unique device number

Example:

E8:BA:70 : 11:28:74
| OUI | | Device ID |

- **OUI** → Assigned to companies (Cisco, Intel, HP, etc.)
- **Device ID** → Unique number for each device produced by that company

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🏢 Daily Life Example

Think of a **MAC Address like an Aadhaar Number** 🆔

- Every person has **one unique Aadhaar number**
- Every network device has **one unique MAC address**

So inside a LAN, devices recognize each other using **MAC addresses**.

🔹 2️⃣ Communication in a Switch Network

A **switch works at Layer 2 of the OSI model**.

📌 Switch uses **MAC addresses** to send data between devices.

Important point:

✔ Switch **does NOT need IP addresses** to forward frames inside a LAN.

Instead it uses:

MAC Address

📘 3️⃣ Switch MAC Address Table

🧠 What is a MAC Address Table?

A **MAC Address Table** is a table inside a switch that stores:

- Device **MAC address**
- **Interface/port** where the device is connected

Example table:

MAC Address	Interface
AA:BB:CC:11:22:33	Fa0/1
DD:EE:FF:44:55:66	Fa0/3

This allows the switch to send frames **directly to the correct device**.

🔍 How Switch Learns MAC Addresses

Switch learns MAC addresses using:

SOURCE MAC ADDRESS

When a frame arrives:

1️⃣ Switch checks **source MAC address**
2️⃣ Switch records it in **MAC table**
3️⃣ Associates it with the **incoming port**

📡 Example: Communication Between PCs

Imagine a switch connected to **4 PCs**.

PC1
PC2
PC3
PC4

PC1 wants to communicate with **PC3**.

Step 1 – PC1 sends frame

PC1 sends a frame asking:

Who has MAC address of PC3?

Switch does **not know PC3 yet**.

🚀 Unknown Unicast Frame

When the switch **doesn't know the destination MAC**, it sends the frame to **all ports except the incoming port**.

This process is called:

UNKNOWN UNICAST FLOODING

Example:

PC1 connected to **Fa0/1**

Switch forwards frame to:

Fa0/2
Fa0/3
Fa0/4

But **not Fa0/1**.

📩 Response from PC3

When **PC3 receives the frame**:

✔ It sends a reply to PC1.

Now the switch learns:

MAC	Interface
PC1	Fa0/1
PC3	Fa0/3

Now communication becomes **direct**.

🎯 Known Unicast Frame

Now the switch knows where PC3 is.

Next time PC1 sends traffic:

Switch sends frame ONLY to Fa0/3

This is called:

KNOWN UNICAST FORWARDING

⏱️ MAC Address Aging

MAC table entries are **temporary**.

If no traffic comes from a device for **5 minutes**, the switch removes the entry.

MAC Aging Timer = 300 seconds

📌 Reason:

- Prevents old or invalid entries
- Keeps MAC table updated

💻 Commands to View and Manage MAC Table

1️⃣ Show MAC Address Table

show mac address-table

Shows:

- All learned MAC addresses
- Corresponding switch ports

2️⃣ Show MAC Table of Specific Interface

show mac address-table interface fa0/1

Shows MAC addresses learned only on **Fa0/1**.

3️⃣ Delete All Dynamic MAC Entries

clear mac address-table dynamic

Deletes all **dynamic entries**.

❗ Static entries remain.

4️⃣ Delete MAC Entries of Specific Interface

clear mac address-table dynamic interface fa0/1

Deletes entries only for **Fa0/1**.

5️⃣ Delete Specific MAC Address

clear mac address-table dynamic address 0001.42ab.23cd

Removes one specific MAC entry.

🌐 4️⃣ What is a LAN (Local Area Network)?

A **LAN** is a **private network** where devices communicate within the same area.

Examples:

🏢 Office network
🏫 School network
🏠 Home WiFi network

Devices inside a LAN can share:

- Files
- Printers
- Internet
- Applications

📡 Broadcast Domain

A **broadcast domain** is a group of devices that receive a **broadcast frame**.

Broadcast MAC address:

FFFF.FFFF.FFFF

Meaning:

Send to ALL devices in the network

⚠️ Problem with Large Broadcast Domains

Too many broadcasts cause:

🐌 Performance Issues

Unnecessary traffic reduces network speed.

🔓 Security Issues

Devices can communicate freely without restrictions.

Example:

One employee accessing **confidential HR systems**.

🔹 5️⃣ VLAN (Virtual Local Area Network)

📘 What is a VLAN?

A **VLAN** is a logical separation of devices inside a switch.

Even if devices are connected to the same switch, VLANs divide them into **separate networks**.

📌 VLANs work at **Layer 2**.

🧠 Example

Office departments:

Engineering
HR
Sales

Instead of installing **3 physical networks**, we create **3 VLANs**.

VLAN	Department
VLAN 10	Engineering
VLAN 20	HR
VLAN 30	Sales

Devices in VLAN 10 **cannot communicate directly** with VLAN 20.

🎯 Purpose of VLANs

🚀 Improve Network Performance

Broadcast traffic stays **inside the VLAN only**.

This reduces congestion.

🔐 Improve Security

Devices from other VLANs cannot access data.

Example:

Engineering PCs cannot access HR systems.

🔁 Communication Between VLANs

Switch **cannot route between VLANs**.

For communication between VLANs we need:

Router

This process is called:

INTER-VLAN ROUTING

Methods:

- Router on a Stick
- Layer 3 Switch

🔌 Switch Ports

💻 Access Port

An **Access Port** belongs to **only one VLAN**.

Used to connect:

- PCs
- Printers
- Servers

Example:

switchport mode access

🌉 Trunk Port

A **Trunk Port** carries **multiple VLANs**.

Used between:

- Switch ↔ Switch
- Switch ↔ Router

🧪 VLAN LAB – Packet Tracer

🗺️ Network Design

VLAN	Name	Network	Gateway
10	Engineering	10.0.0.0/26	10.0.0.62
20	HR	10.0.0.64/26	10.0.0.126
30	Sales	10.0.0.128/26	10.0.0.190

🖥️ PC IP Configuration

PC	IP	Gateway
PC1	10.0.0.1	10.0.0.62
PC2	10.0.0.2	10.0.0.62
PC3	10.0.0.65	10.0.0.126
PC4	10.0.0.66	10.0.0.126
PC5	10.0.0.129	10.0.0.190
PC6	10.0.0.130	10.0.0.190

Subnet Mask for all PCs:

255.255.255.192

🔧 Router Configuration

```
R1> enable
R1# configure terminal

interface g0/0
ip address 10.0.0.62 255.255.255.192
no shutdown
```

interface g0/1
ip address 10.0.0.126 255.255.255.192
no shutdown

interface g0/2
ip address 10.0.0.190 255.255.255.192
no shutdown

⚙️ Switch VLAN Configuration

Create VLANs

SW1> enable
SW1# configure terminal

vlan 10
name Engineering

vlan 20
name HR

vlan 30
name Sales

🔌 Assign Ports to VLANs

interface fa3/1
switchport mode access
switchport access vlan 10

interface fa4/1
switchport mode access
switchport access vlan 10

interface fa5/1
switchport mode access
switchport access vlan 20

```
interface fa6/1
switchport mode access
switchport access vlan 20

interface fa7/1
switchport mode access
switchport access vlan 30

interface fa8/1
switchport mode access
switchport access vlan 30
```

🔍 Verify VLANs

```
show vlan brief
```

Shows:

- VLAN numbers
- VLAN names
- Ports assigned

📡 Testing Connectivity

Normal Ping

```
PC1 ping 10.0.0.2
PC1 ping 10.0.0.65
```

📢 Broadcast Ping

Engineering VLAN:

```
PC1 ping 10.0.0.63
```

HR VLAN:

```
PC3 ping 10.0.0.127
```

Sales VLAN:

PC5 ping 10.0.0.191

🎮 Packet Tracer Simulation Mode

Steps:

1️⃣ Select **Simulation Mode**
2️⃣ Enable **ICMP Filter**
3️⃣ Send Broadcast Ping
4️⃣ Click **Capture/Forward**

You will see the **packet flow**.

📊 Expected Result

If Engineering PC sends broadcast:

Device	Receive Broadcast
Engineering PCs	✅ Yes
HR PCs	❌ No
Sales PCs	❌ No

Reason:

Each VLAN = Separate Broadcast Domain

🌍 Real Life Example

In a company:

🏢 Engineering department
🏢 HR department
🏢 Sales department

Each department has:

- Separate VLAN
- Separate broadcast traffic
- Better security

But all use **same physical switch infrastructure**.

🎯 One-Liners

✓ MAC Address

"A MAC address is a 48-bit unique hardware identifier assigned to a network interface card."

✓ VLAN

"A VLAN logically separates devices into different broadcast domains on the same physical switch."

🧠 Final Key Points

✓ MAC Address = Unique device identifier
✓ Switch learns MAC using **source address**
✓ Unknown unicast = **Flooding**
✓ Known unicast = **Forwarding**
✓ MAC table aging = **5 minutes**
✓ VLAN = Logical network separation
✓ VLAN improves **performance & security**
✓ Each VLAN = **Separate broadcast domain**

📌 1️⃣ Chapter Summary 🧠

🔹 MAC Address

- A **MAC Address (Media Access Control Address)** is a **unique hardware identifier** assigned to a **Network Interface Card (NIC)**.
- It is **48 bits long (6 bytes)** and written in **hexadecimal format**.
- Example:
  E8:BA:70:11:28:74

Structure:

Part	Size	Purpose
OUI	First 3 bytes	Manufacturer identifier
Device ID	Last 3 bytes	Unique device number

📌 Used for **communication inside LAN networks**.

🔹 Switch MAC Address Table

A **MAC Address Table** is maintained by a switch that maps:

MAC Address	Port
Device MAC	Interface where device is connected

The switch **learns MAC addresses dynamically** using the **source MAC address** of incoming frames.

🔹 Unknown Unicast Flooding

If the switch **does not know the destination MAC**, it sends the frame to **all ports except the incoming port**.

This is called:

📡 **Unknown Unicast Flooding**

🔹 Known Unicast Forwarding

Once the switch learns the MAC address:

✔ It sends the frame **only to the correct port**.

This improves **network efficiency**.

🔹 MAC Address Aging

- Switch removes unused MAC entries after **300 seconds (5 minutes)**.
- This keeps the MAC table **updated and efficient**.

🔹 LAN (Local Area Network)

A **LAN** is a network where devices communicate within a **limited geographic area**.

Examples:

🏠 Home network
🏢 Office network
🏫 School network

Devices share:

- Files
- Printers
- Internet
- Applications

🔹 Broadcast Domain

A **broadcast domain** is a group of devices that receive **broadcast frames**.

Broadcast MAC Address:

FFFF.FFFF.FFFF

Meaning:
📢 Send packet to **all devices in the network**.

🔹 VLAN (Virtual LAN)

A **VLAN** logically separates devices on the same switch into **different networks**.

Example:

VLAN	Department
VLAN 10	Engineering
VLAN 20	HR
VLAN 30	Sales

Benefits:

🚀 Better performance
🔐 Better security
📉 Reduced broadcast traffic

🔹 Inter-VLAN Routing

Devices in different VLANs **cannot communicate directly**.

Communication requires:

- Router
- Layer 3 Switch

This process is called:

🔁 **Inter-VLAN Routing**

🔹 Switch Ports

Access Port

Connects end devices.

Example command:

```
switchport mode access
switchport access vlan 10
```

Trunk Port

Carries traffic from **multiple VLANs**.

Used between:

- Switch ↔ Switch
- Switch ↔ Router

🎯 Chapter Conclusion

This chapter explains the **foundation of Layer-2 networking**.

Key ideas:

✔ Every device in a network has a **unique MAC address**
✔ Switches use **MAC tables** to forward frames efficiently
✔ Unknown destinations cause **frame flooding**
✔ MAC entries expire using the **aging timer**
✔ **LANs** allow devices to communicate locally
✔ **Broadcast domains** affect performance in large networks
✔ **VLANs logically divide networks** for security and efficiency
✔ **Inter-VLAN routing** allows communication between VLANs

📌 In real enterprise networks, **VLAN segmentation is one of the most important security and performance techniques.**

🎤 Q & A

1️⃣ What is a MAC Address? 🧠

Answer

A **MAC Address (Media Access Control Address)** is a **unique hardware identifier assigned to a network interface card (NIC)**.

Key features:

- 48-bit address
- Written in hexadecimal
- Stored permanently in NIC
- Also called **BIA (Burned-In Address)**

Example:

E8:BA:70:11:28:74

It is used by **Layer 2 devices like switches** to identify devices inside a **Local Area Network (LAN)**.

📌 Simple explanation:

Think of a MAC address like a **device fingerprint** inside the network.

2️⃣ What is the structure of a MAC Address?

A MAC address has **two parts**.

Part	Description
OUI	Organizationally Unique Identifier
Device ID	Unique number for device

Example:

E8:BA:70 : 11:28:74

- First part identifies manufacturer

- Second part identifies device

Companies like **Cisco, Intel, HP** receive OUIs from IEEE.

3️⃣ What is a MAC Address Table in a switch?

Answer

A **MAC Address Table** is a table maintained by a switch that stores:

- MAC address
- Associated switch port

Example:

MAC Address	Interface
AA:BB:CC:11:22:33	Fa0/1

This allows the switch to **forward frames efficiently**.

4️⃣ How does a switch learn MAC addresses?

Switches learn MAC addresses using the **Source MAC Address** of incoming frames.

Steps:

1️⃣ Frame arrives at switch
2️⃣ Switch reads source MAC
3️⃣ Stores MAC in table
4️⃣ Associates it with incoming port

This process is called:

📡 **MAC Learning**

5️⃣ What is Unknown Unicast Flooding?

Answer

Unknown unicast flooding occurs when:

- Switch receives a frame
- Destination MAC is **not present in MAC table**

Switch sends the frame to **all ports except the incoming port**.

Purpose:

✓ Find the destination device.

6️⃣ What is Known Unicast Forwarding?

Answer

When the switch **knows the destination MAC address**, it forwards the frame only to the correct port.

Benefits:

✓ Faster communication
✓ Less network traffic

7️⃣ What is MAC Address Aging?

Answer

MAC table entries are **temporary**.

If a device does not send traffic for **300 seconds (5 minutes)**, the switch removes the entry.

Purpose:

✓ Prevent outdated entries
✓ Maintain table efficiency

8️⃣ What is a LAN?

A **Local Area Network (LAN)** is a network where devices communicate within a **limited geographic area**.

Examples:

🏠 Home network
🏢 Office network
🏫 School network

LANs typically use:

- Ethernet
- WiFi

9️⃣ What is a Broadcast Domain?

A **broadcast domain** is a network segment where broadcast packets reach **all devices**.

Broadcast address:

FFFF.FFFF.FFFF

Problems with large broadcast domains:

🐌 Network congestion
🔓 Security risks

🔟 What is a VLAN?

A **VLAN (Virtual Local Area Network)** logically separates devices on a switch into **different broadcast domains**.

Example:

VLAN	Department
10	Engineering
20	HR

VLAN	Department
30	Sales

Benefits:

✓ Security
✓ Traffic isolation
✓ Better performance

1️⃣1️⃣ Why VLANs are used?

Main reasons:

🚀 Reduce broadcast traffic
🔐 Improve security
📈 Improve performance
🏢 Organize departments

1️⃣2️⃣ Can devices in different VLANs communicate?

No.

Devices in different VLANs require:

🔁 **Inter-VLAN Routing**

This can be done using:

- Router on a Stick
- Layer 3 Switch

1️⃣3️⃣ What is an Access Port?

An **Access Port** belongs to a single VLAN.

Used for:

- PCs
- Printers

- Servers

Example command:

```
switchport mode access
switchport access vlan 10
```

1️⃣4️⃣ What is a Trunk Port?

A **Trunk Port** carries traffic for multiple VLANs.

Used between:

- Switch ↔ Switch
- Switch ↔ Router

It uses **VLAN tagging**.

1️⃣5️⃣ What command shows MAC table?

```
show mac address-table
```

Displays:

- Learned MAC addresses
- Associated ports
- VLAN information

1️⃣6️⃣ How do you clear MAC table?

Delete all entries:

```
clear mac address-table dynamic
```

Delete specific interface entries:

```
clear mac address-table dynamic interface fa0/1
```

Delete specific MAC:

```
clear mac address-table dynamic address 0001.42ab.23cd
```

1️⃣7️⃣ What command verifies VLAN configuration?

show vlan brief

Displays:

- VLAN ID
- VLAN name
- Assigned ports

1️⃣8️⃣ Real-World Interview Scenario Question

Question

A user from HR cannot communicate with Engineering PC. Why?

Answer

Because:

- HR PC is in **VLAN 20**
- Engineering PC is in **VLAN 10**

Different VLANs cannot communicate without **Inter-VLAN routing**.

Solution:

Configure:

- Router
- Layer 3 Switch

1️⃣9️⃣ What happens when a broadcast packet is sent?

Broadcast packets are delivered to:

✓ All devices in same VLAN

Devices in other VLANs:

❌ Do not receive it.

2️⃣0️⃣ What are advantages of VLAN?

Major benefits:

✓ Reduced broadcast traffic
✓ Improved security
✓ Better network organization
✓ Logical segmentation

⭐ "Explain VLAN in one line."

Answer:

"A VLAN logically separates devices into different broadcast domains on the same physical switch to improve network security and performance."

🎯 Final Important Points to Remember

✓ MAC Address = Unique device identifier
✓ Switch learns MAC from **source address**
✓ Unknown unicast causes **flooding**
✓ Known unicast allows **direct forwarding**
✓ MAC aging timer = **300 seconds**
✓ VLAN divides network into **multiple broadcast domains**
✓ Inter-VLAN routing allows VLAN communication
